ServiceNow + Microsoft Copilot AI Incident Routing 系统

整体架构设计方案

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1. 执行摘要

1.1 项目背景

当前ServiceNow incident分配依赖于静态规则或人工判断，无法充分利用KB中的处理流程知识。本项目旨在构建一个AI驱动的智能路由系统，能够： - 根据CI关联的KB自动推荐assignment group - 通过Copilot提供自然语言交互界面 - P1 incident自动触发通知机制

1.2 核心目标

目标	描述
智能路由	基于KB内容自动推荐assignment group
AI助手	通过Copilot提供交互式建议查询
快速响应	P1 incident自动触发多渠道通知
知识整合	RAG技术让AI学习企业KB内容

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2. 系统架构总览

2.1 整体架构图

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                              用户交互层 (Presentation Layer)                  │
```

```
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────────────────────┐ │
│  │ ServiceNow   │  │ Microsoft    │  │ Virtual Agent / Now Assist   │ │
│  │ Incident     │  │ Teams        │  │ (自然语言界面)                 │ │
│  │ Form         │  │ Copilot      │  │                              │ │
│  └──────┬───────┘  └──────┬───────┘  └──────────────┬───────────────┘ │
└─────────┼─────────────────┼─────────────────────────┼─────────────────┘
          │                 │                         │
          ▼                 ▼                         ▼

┌──────────────────────────────────────────────────────────────────────┐
│                      业务逻辑层 (Business Logic Layer)                 │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────────────────┐   │
│  │ Business Rule │  │ AI Routing   │  │ Notification Engine      │   │
│  │ (触发逻辑)     │  │ Engine       │  │ (P1 Alert)               │   │
│  └──────┬───────┘  └──────┬───────┘  └──────────┬───────────────┘   │
│         │                 │                     │                   │
│  ┌──────▼───────┐  ┌──────▼───────┐  ┌──────────▼───────────────┐   │
│  │ Flow Designer │  │ Script       │  │ On-Call Schedule         │   │
│  │ (Workflow)    │  │ Include      │  │ Integration              │   │
│  └──────────────┘  └──────────────┘  └──────────────────────────┘   │
└──────────────────────────────────────────────────────────────────────┘
          │
          ▼

┌──────────────────────────────────────────────────────────────────────┐
│                      AI/ML 层 (AI Service Layer)                      │
│  ┌──────────────────────────────────────────────────────────────┐    │
│  │              Microsoft Azure OpenAI Service                   │    │
│  │  ┌──────────────┐  ┌──────────────┐  ┌──────────────────┐   │    │
│  │  │ GPT-4/GPT-4o │  │ RAG Engine   │  │ Embeddings       │   │    │
│  │  │ (LLM)        │  │ (KB检索)      │  │ (向量化)          │   │    │
│  │  └──────────────┘  └──────────────┘  └──────────────────┘   │    │
│  └──────────────────────────────────────────────────────────────┘    │
│                              ▲                                       │
│                              │                                       │
└──────────────────────────────┼───────────────────────────────────────┘
                               │
                               ▼

┌──────────────────────────────────────────────────────────────────────┐
│                      数据层 (Data Layer)                               │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐  ┌──────────┐ │
│  │ Incident     │  │ KB Articles  │  │ CMDB (CI)    │  │ AI Search │ │
│  │ Table        │  │ (处理流程)     │  │ (support_grp)│  │ Index     │ │
```

```
│   └──────────────┘   └──────────────┘   └──────────────┘   └──────────────┘   │
└──────────────────────────────────────────────────────────────────────────────┘
```

2.2 数据流架构

```
Incident Creation Flow:
═══════════════════════════════════════════════════════════════════════════

User creates Incident
        │
        ▼
┌──────────────────┐
│ Business Rule    │──[1]──▶ Check if cmdb_ci is set
│ (After Insert)   │
└──────────────────┘
         │
         ▼
┌──────────────────┐
│ KB Retrieval     │──[2]──▶ Query KB articles related to CI
│ (RAG - Step 1)   │         (use AI Search or scripted query)
└──────────────────┘
         │
         ▼
┌──────────────────┐
│ AI Analysis      │──[3]──▶ Send KB content + Incident details
│ (Azure OpenAI)   │         to LLM for assignment recommendation
└──────────────────┘
         │
         ▼
┌──────────────────┐
│ Parse Response   │──[4]──▶ Extract assignment_group from AI response
│ (Script Include)│
└──────────────────┘
         │
         ▼
┌──────────────────┐
│ Update Incident  │──[5]──▶ Set assignment_group
│ (Set Values)     │         Add work note with AI reasoning
└──────────────────┘
         │
```

```
        ▼
┌──────────────────┐
│ Priority Check   │──[6]──▶ If Priority = 1 (P1)
│ (Condition)      │         Trigger notification workflow
└──────────┬───────┘
           │
     ┌─────┴─────┐
     ▼           ▼
   P1=Y        P1=N
     │           │
     ▼           │
┌──────────────────┐
│ P1 Notification  │──[7]──▶ Send SMS/Email/Teams notification
│ Workflow         │         to on-call engineer + manager
└──────────────────┘

Copilot Interaction Flow:
═══════════════════════════════════════════════════════════════════

Support Engineer opens Teams Copilot
        │
        ▼
┌──────────────────┐
│ Natural Lang     │──[8]──▶ "Who should handle this incident?"
│ Query Input      │
└──────────┬───────┘
           │
           ▼
┌──────────────────┐
│ Copilot routes   │──[9]──▶ Intent recognition (Now Assist NLU)
│ to Now Assist    │
└──────────┬───────┘
           │
           ▼
┌──────────────────┐
│ Virtual Agent    │──[10]──▶ Fetch incident details
│ Topic Execution  │         Query KB for similar cases
└──────────┬───────┘
           │
           ▼
┌──────────────────┐
```

```
│ AI generates     │──[11]──▶ Azure OpenAI generates recommendation
│ recommendation   │         with explanation
└──────────┬───────┘
           │
           ▼
┌──────────────────┐
│ Display in       │──[12]──▶ Show recommendation + confidence score
│ Teams Chat       │         + option to apply assignment
└──────────────────┘
```

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3. 核心组件设计

3.1 CI-KB关联模型

```
// CI-KB关联数据结构
{
    "cmdb_ci": {
        "sys_id": "unique_ci_id",
        "name": "SAP Production Server",
        "support_group": "SAP Team (sys_id)",
        "kb_articles": [
            {
                "number": "KB0012345",
                "title": "SAP Server Incident Handling",
                "category": "escalation_procedure",
                "assignment_rules": {
                    "network_issue": "Network Team",
                    "db_issue": "DBA Team",
                    "app_issue": "SAP Team"
                }
            }
        ]
    }
}
```

3.2 AI路由引擎设计

```
// Script Include: AIRoutingEngine
var AIRoutingEngine = Class.create();
AIRoutingEngine.prototype = {
    initialize: function() {
        this.azureEndpoint = gs.getProperty('azure.openai.endpoint');
        this.apiKey = gs.getProperty('azure.openai.key');
        this.deploymentName = gs.getProperty('azure.openai.deployment');
    },

    // 主函数：获取assignment group推荐
    getAssignmentRecommendation: function(incidentId) {
        var incident = new GlideRecord('incident');
        if (!incident.get(incidentId)) return null;

        // 1. 收集上下文信息
        var context = this._gatherContext(incident);

        // 2. 检索相关KB
        var kbContent = this._retrieveKBContent(context.ciSysId);

        // 3. 构建prompt
        var prompt = this._buildPrompt(incident, context, kbContent);

        // 4. 调用Azure OpenAI
        var aiResponse = this._callAzureOpenAI(prompt);

        // 5. 解析响应
        return this._parseAIResponse(aiResponse);
    },

    // RAG: 检索增强生成
    _retrieveKBContent: function(ciSysId) {
        // 使用ServiceNow AI Search或直接查询
        var kbGR = new GlideRecord('kb_knowledge');
        kbGR.addQuery('cmdb_ci', ciSysId);
        kbGR.addActiveQuery();
        kbGR.query();
```

```
        var articles = [];
        while (kbGR.next()) {
            articles.push({
                number: kbGR.getValue('number'),
                title: kbGR.getValue('short_description'),
                content: kbGR.getValue('text')
            });
        }
        return articles;
    },

    _buildPrompt: function(incident, context, kbContent) {
        return {
            system: "You are an expert ITSM routing assistant. Based on the incident d
            user: {
                incident: {
                    short_description: incident.short_description,
                    description: incident.description,
                    category: incident.category,
                    subcategory: incident.subcategory,
                    ci_name: context.ciName,
                    urgency: incident.urgency,
                    impact: incident.impact
                },
                kb_articles: kbContent,
                available_groups: context.availableGroups
            },
            output_format: {
                assignment_group: "recommended_group_name",
                confidence: "high/medium/low",
                reasoning: "explanation"
            }
        };
    }
};
```

3.3 P1通知工作流

```
P1 Notification Workflow:
┌──────────────────┐
```

```
│ Trigger:          │
│ Priority = 1      │
│ AND               │
│ State = New       │
└───────────────────┘
          │
          │
          ▼

┌───────────────────┐
│ Action 1:         │
│ Get On-Call       │
│ Engineer          │
│ (from Schedule)   │
└───────────────────┘
          │
          │
          ▼

┌───────────────────┐
│ Action 2:         │
│ Send Email        │
│ (Manager DL)      │
└───────────────────┘
          │
          │
          ▼

┌───────────────────┐
│ Action 3:         │
│ Send SMS          │
│ (On-Call Phone)   │
└───────────────────┘
          │
          │
          ▼

┌───────────────────┐
│ Action 4:         │
│ Teams Message     │
│ (Escalation Ch)   │
└───────────────────┘
```

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4. Microsoft Copilot 集成方案

4.1 集成架构

组件	功能	配置要点
Custom Engine Agent	Copilot与ServiceNow的桥梁	Zurich+版本内置支持
Now Assist	ServiceNow的AI平台	需Now Assist许可证
Generative AI Controller	连接Azure OpenAI	配置API endpoint和key
Virtual Agent	对话界面	Topic设计和NLU训练

4.2 Copilot交互场景

场景1: Assignment建议查询

```
User: "Who should handle INC0012345?"
Copilot: [识别intent] → 调用Now Assist
Now Assist: [查询incident] → 获取CI和KB信息
Now Assist: [调用Azure OpenAI] → 生成推荐
Copilot: "Based on the KB article KB0012345, this SAP server issue
          should be assigned to the SAP Team.
          Confidence: High
          Would you like me to update the assignment?"
```

场景2: 历史案例查询

```
User: "Have we seen this issue before?"
Copilot: [搜索similar incidents] → 返回历史案例
Copilot: "Found 3 similar incidents in the past 30 days.
          Resolution: Restart SAP service."
```

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5. 实施路线图

5.1 阶段划分

阶段	时间	交付物	负责方
Phase 1	Week 1-2	基础配置：CI-KB关联、Assignment规则	@shirmpclaw
Phase 2	Week 3-4	Azure OpenAI集成、Script Include开发	@shirmpclaw
Phase 3	Week 5-6	P1通知工作流、On-call schedule集成	@shirmpclaw
Phase 4	Week 7-8	Copilot集成、Virtual Agent配置	@shirmpclaw
Phase 5	Week 9-10	测试优化、培训、上线	联合

5.2 依赖关系

```
[CI数据清理] ──────┐
                   ▼
[KB标准化] ──────► [AI Routing Engine] ──────► [Copilot Integration]
                   ▲                            ▲
[Azure OpenAI] ────┘                            │
                                                │
[P1 Notification] ──────────────────────────────┘
```

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6. 风险评估与缓解

风险	影响	缓解措施
AI推荐不准确	高	保留人工覆盖机制，设置confidence阈值
Azure OpenAI延迟	中	异步处理，设置timeout，fallback到静态规则
P1通知失败	高	多渠道冗余(SMS+Email+Teams)
数据隐私	高	PII脱敏，限制AI访问范围

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7. 附录

附录A: 需要的ServiceNow插件

- Virtual Agent
- Now Assist
- Conversational Integration with Microsoft Teams
- Integration Hub (for P1 notifications)

附录B: Azure资源需求

- Azure OpenAI Service (GPT-4/GPT-4o)
- Azure AI Search (for RAG indexing)
- (可选) Azure Functions (for webhook处理)

文档生成时间: 2026-03-08 版本: v1.0 作者: 富贵虾 (AI Architect) + @shirmpclaw (Implementation)